LECTURE NOTES

Lecture 14

AI504 - Knowledge Representation

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May/2026

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Contents

1 quick exercise

Formalize the following statement.

- There are infinity many prime numbers

Lets work in $\mathbb{N}$ Assume then that we have that

`prime(n), <`

$\forall n \in \mathbb{N}, \exists m \in \mathbb{N} : n < m \land \text{prime}(m)$

- define $\text{prime}(n)$

$$\text{prime}(n) \iff n > 1 \land \forall x \in \mathbb{N}.x \mid n \Rightarrow (x = 1 \lor x = n)$$

There are many legitimate ways to express this

$$\text{prime}(n) \iff \forall a, b \in \mathbb{N}.\ (n|a \cdot b \Rightarrow (n|a \lor n|b))$$

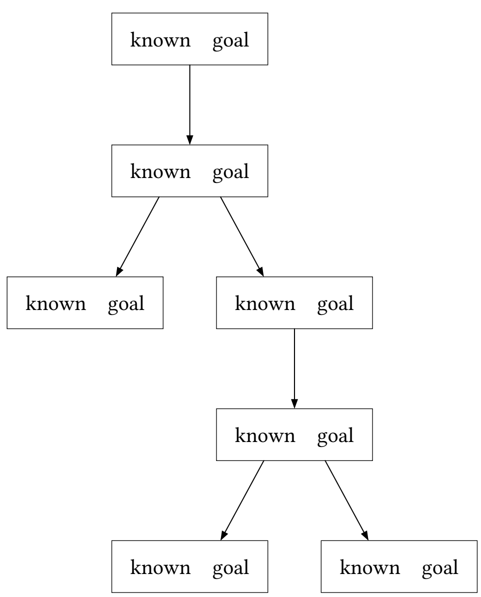
2 The proof theory of first-order logic

- Many different styles of proof calculi
- Here is one, described somewhat informally

A proof in first-order logic is a tree, each node in the tree is labeled by two sets of formulas.

What we know and what are the goals

At each node we pick a formula (either known/goal) and simplify according to the following rules, the other formulas are inherited from above.



Rules

Let the goal be $P \Rightarrow Q$ Then we would Assume P and the new goal is Q

$$\begin{tikzcd}
\boxed{\text{known} : \underline{\qquad} \quad \text{goal} : (P \to Q)} \arrow[d] \\
\boxed{\text{known} : P \quad \text{goal} : Q}
\end{tikzcd}$$

Let $P, P \Rightarrow Q$ be known, then

$$\begin{tikzcd}
\boxed{\text{known} : P, P \Rightarrow Q \quad \text{goal} : \underline{\qquad}} \arrow[d] \\
\boxed{\text{known} : P, Q \quad \text{goal} : \underline{\qquad}}
\end{tikzcd}$$

Let $P \vee Q$ be known

$$\begin{tikzcd}
& \boxed{\text{known} : P \vee Q \quad \text{goal} : \underline{\qquad}} \arrow[dl] \arrow[dr] & \\
\boxed{\text{known} : P \quad \text{goal} : \underline{\qquad}} & & \boxed{\text{known} : Q \quad \text{goal} : \underline{\qquad}}
\end{tikzcd}$$

Let $P \wedge Q$ be the goal

$$\begin{tikzcd}
\boxed{\text{known} : \underline{\qquad} \quad \text{goal} : P \wedge Q} \arrow[d] \\
\boxed{\text{known} : \underline{\qquad} \quad \text{goal} : P, Q}
\end{tikzcd}$$

Let $P \vee Q$ be the goal

$$\begin{tikzcd}
\boxed{\text{known} : \underline{\qquad} \quad \text{goal} : P \vee Q} \arrow[d] & \boxed{\text{known} : \underline{\qquad} \quad \text{goal} : P \vee Q} \arrow[d] \\
\boxed{\text{known} : \underline{\qquad} \quad \text{goal} : P} & \boxed{\text{known} : \underline{\qquad} \quad \text{goal} : Q}
\end{tikzcd}$$

Let $P \wedge Q$ be known

$$\begin{tikzcd} \boxed{\text{known} : P \wedge Q \quad \text{goal} : \_\_\_\_\_\_} \arrow[d] & \boxed{\text{known} : P \wedge Q \quad \text{goal} : \_\_\_\_\_\_} \arrow[d] \\ \boxed{\text{known} : P \quad \text{goal} : \_\_\_\_\_\_} & \boxed{\text{known} : Q \quad \text{goal} : \_\_\_\_\_\_} \end{tikzcd}$$

$\forall$ is known. “for all x, x has property ϕ”

$$\begin{tikzcd} \boxed{\forall x \phi} \arrow[d] \\ \boxed{\underbrace{\phi(t)}_{\text{term } t \text{ for x}}} \end{tikzcd}$$

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$\exists$ is known. “there exists an x, such that x has property ϕ”

$$\begin{tikzcd} \boxed{\exists x \phi} \arrow[d] \\ \boxed{\underbrace{\phi(t)}_{\text{term } t \text{ for x}}} \end{tikzcd}$$

$\exists$ is goal. “there exists an x, such that x has property ϕ”

$$\begin{tikzcd} \boxed{\exists x \phi} \arrow[d] \\ \boxed{\phi} \end{tikzcd}$$